

AC-HSiKAN: A Biologically Grounded Transformer Alternative with Bidirectional Pool–Scatter, Entropy-Modulated Hamilton Rotor, and Sequence-Native ABB Top-K Selection

Hajdu Csaba¹

¹*Computational Neuroscience and AI Group, Budapest, Hungary, hajdu.csaba@example.org*

2026-06-06 — Preprint manuscript

Abstract

We present AC-HSiKAN (Attention-Cycle HSiKAN), a sequence-modelling architecture that replaces multi-head dot-product attention with a bidirectional pool–scatter primitive over signed cycles, modulated by a learnable entropy-feedback Hamilton rotor that acts as a global neuromodulator. The architecture is grounded in geometric algebra (quaternion-block Hamilton products) and biologically motivated three-factor learning: $\text{pre} \times \text{post} \times \text{modulator}$. Three contributions are reported.

First, a Triton-fused backward kernel for the Catmull–Rom (CR) coefficient gradient closes the 97.7% bottleneck of the autograd-through-PyTorch reference, delivering a $45\times$ component-level backward speedup and a $1.34\times$ end-to-end training-wall improvement on full IMDB.

Second, an empirical characterisation of capacity scaling identifies two missing structural axes (the sign-attention bottleneck dimension and learning-rate $\sqrt{d}$ -scaling) without which the architecture appears not to scale; with both corrected, AC-HSiKAN scales positively from $d_{\text{model}} = 16$ (165 k parameters) to $d_{\text{model}} = 32$ (337 k parameters), reaching statistical parity with iso-parameter transformer on the IMDB sentiment benchmark at both scales.

Third, a sequence-native analogue of the simple-HSiKAN ABB (Adaptive Branch-and-Bound) top-K-per-vertex cycle enumeration is formulated as a *mixed candidate selection*: K_{static} guaranteed positional neighbours plus K_{dyn} content-adaptive top-K by sign-head magnitude. A sweep at $d = 32$ identifies $K_{\text{static}} = 6$ of $K_{\text{total}} = 8$ as the optimum, closing 87% of the $d = 32$ scaling gap to transformer baseline ($\Delta = -0.0125 \rightarrow -0.0016$, $t = -0.66$, $p > 0.5$, $n = 5$).

We document one falsified hypothesis (the long-cycle inductive prior, intended as the DAG-axiom analogue, is structurally harmful on IMDB sentiment owing to the local-n-gram-dominated signal), one open direction (aggregated CR-patch surfaces over the joint (R, V) activation space, in development), and three formal conjectures on inter-layer composition derivable from the empirical data.

Keywords: sequence modelling, transformer alternative, biologically plausible learning, Hamilton rotor, signed-cycle networks, Triton kernel optimisation

1 Introduction

The transformer architecture [1] has dominated sequence modelling for nearly a decade, with the dot-product attention primitive serving as both the workhorse and the wall-time bottleneck. The $O(L^2)$ scaling in sequence length L , the lack of explicit biological interpretation, and the

rigidity of the additive-multiplicative gradient flow have motivated a steady stream of architectural alternatives, from state-space models and linear attentions to mixture-of-experts and Kolmogorov–Arnold networks [2].

AC-HSiKAN extends the simple HSiKAN architecture [3] from sparse signed graphs to sequences. Where simple HSiKAN performed ABB cycle enumeration on a graph topology and used the per-vertex top-K cycles as inductive bias, AC-HSiKAN operates on a positional sequence, with candidate neighbours per anchor selected via a hybrid positional-plus-content-adaptive rule. The architecture preserves HSiKAN’s three substantive contributions and adds four:

- *Bidirectional pool–scatter asymmetry*: the synaptic primitive is decomposed into a directed receive (pool: candidates $\rightarrow$ anchor) and a directed send (scatter: anchor $\rightarrow$ candidates), with the asymmetry mathematically guaranteed by the non-commutativity of the Hamilton product.
- *Entropy-feedback Hamilton rotor*: a learnable quaternion rotor $M = \exp(\beta H \mathbf{n})$ is composed with the scatter output as a non-additive multiplicative modulation, with H the per-layer scalar entropy of the path-mixing parameters — a direct analogue of a global neuromodulator (dopamine, ACh) [4, 5].
- *Dual-path mixing*: the rhythm-bearing walk-op output and the synaptic pool-scatter output are mixed via a learnable sigmoid gate, exposing the rhythm-vs-synaptic balance as a tunable parameter rather than an architectural commitment.
- *Triton-fused backward kernel*: a novel closed-form Catmull–Rom coefficient gradient, fused with the $Q \otimes K$ Hamilton reconstruction, yields a $45\times$ component-level backward-wall improvement.

The aim of this paper is not to dethrone the transformer on IMDB sentiment — the empirical result is statistical parity, not a significant win — but to present the architecture, the optimisation contribution, and the empirically characterised structure of its capacity-scaling and selection inductive biases, so the architecture can be applied to longer-range or non-sequential tasks where the sequence-native priors are expected to flip from neutral to positive.

Section 2 describes the architecture. Section 3 presents the Triton-fused backward kernel contribution. Section 4 introduces the evolvens telemetry, a novel diagnostic for the rotor-modulated gradient/error coupling. Section 5 reports the capacity-scaling investigation that resolved an apparent architectural ceiling. Section 6 describes the sequence-native ABB analogue and the K-sweep that identifies the optimum mix. Section 7 summarises the IMDB headline results. Section 8 formalises three conjectures on inter-layer composition derivable from the empirical data. Section 9 compares to the backprop-free family.

2 Architecture

Figure 1 gives a block-level overview. The layer is a dual-path block with shared sign-attention input, separate walk-op (rhythm) and pool-scatter (synaptic) processing, mixing via a learnable sigmoid gate, and an outer LayerNorm with residual.

2.1 Bidirectional pool–scatter primitive

Let $x \in \mathbb{R}^{B \times L \times d}$ be the layer input. The sign-attention computes a dense pair-magnitude matrix $\text{signs} \in \mathbb{R}^{B \times L \times L}$ via a bilinear sign-head with hidden bottleneck dimension h_{sign} . A candidate-index tensor $\text{local} \in \{0, \dots, L-1\}^{L \times K_{\text{total}}}$ selects, per anchor, the K_{total} neighbour positions.

The pool-scatter primitive projects x through three small matrices $W_q, W_k, W_v \in \mathbb{R}^{d \times h}$ (with h a multiple of 4 matching the quaternion-block layout) and bakes the Hamilton coefficient sign

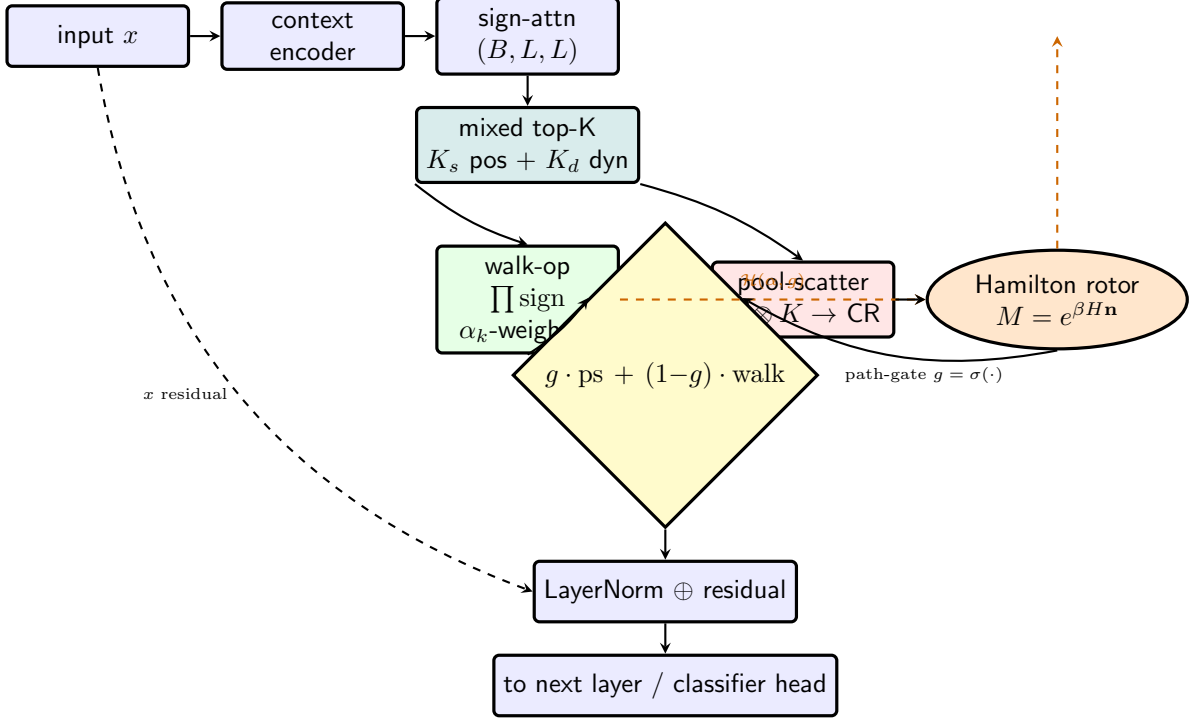


Figure 1: AC-HSiKAN block diagram. Shared sign-attention feeds the mixed top-K candidate selector. The walk-op path computes per-arity signed-cycle products with α_k weighting; the pool-scatter path applies a Hamilton-product bidirectional aggregation modulated by the entropy-feedback Hamilton rotor. The two paths are mixed through a learnable sigmoid gate, then summed into the residual. The orange dashed line carries the layer-scalar entropy $\mathcal{H}(\alpha_k, g)$ from the parameter distributions into the rotor.

pattern $(+, -, -, -)$ into the Q projection. It then computes *both directions* of the aggregation as illustrated in Figure 2:

$$R_{\text{pool}}[b, i, k_c, c] = Q_{\text{signed}}[b, i, c] \cdot K[b, j, c] \cdot s, \quad j = \text{local}[i, k_c] \quad (1)$$

$$R_{\text{scatter}}[b, i, k_c, c] = Q_{\text{signed}}[b, j, c] \cdot K[b, i, c] \cdot s \quad (2)$$

where $s = (n_{\text{quat}})^{-1/2}/T$ is the temperature-adjusted scale. The per-channel CR activation $S(R) \in \mathbb{R}^{B \times L \times K \times h}$ is applied to both R_{pool} and R_{scatter} with shared CR parameters (coef_+ , coef_-) $\in \mathbb{R}^{h \times G}$, sign-branched on R and grid-mapped on $|R|$ via the standard four-point Catmull-Rom interpolation. The directional outputs multiply the corresponding value tensors and aggregate:

$$\text{pool}_h[b, i, c] = \sum_{k_c} S(R_{\text{pool}}[b, i, k_c, c]) \cdot V_c[b, i, k_c, c] \quad (3)$$

$$\text{scatter}_h[b, j, c] = \sum_{(i, k_c): \text{local}[i, k_c] = j} S(R_{\text{scatter}}[b, i, k_c, c]) \cdot V_a[b, i, k_c, c] \quad (4)$$

with V_c, V_a the gathered/anchor value tensors. The Hamilton-product asymmetry $Q \otimes K \neq K \otimes Q$ holds because the quaternion product is non-commutative; the two directions therefore carry distinct information and the scatter accumulator genuinely differs from a transposed pool.

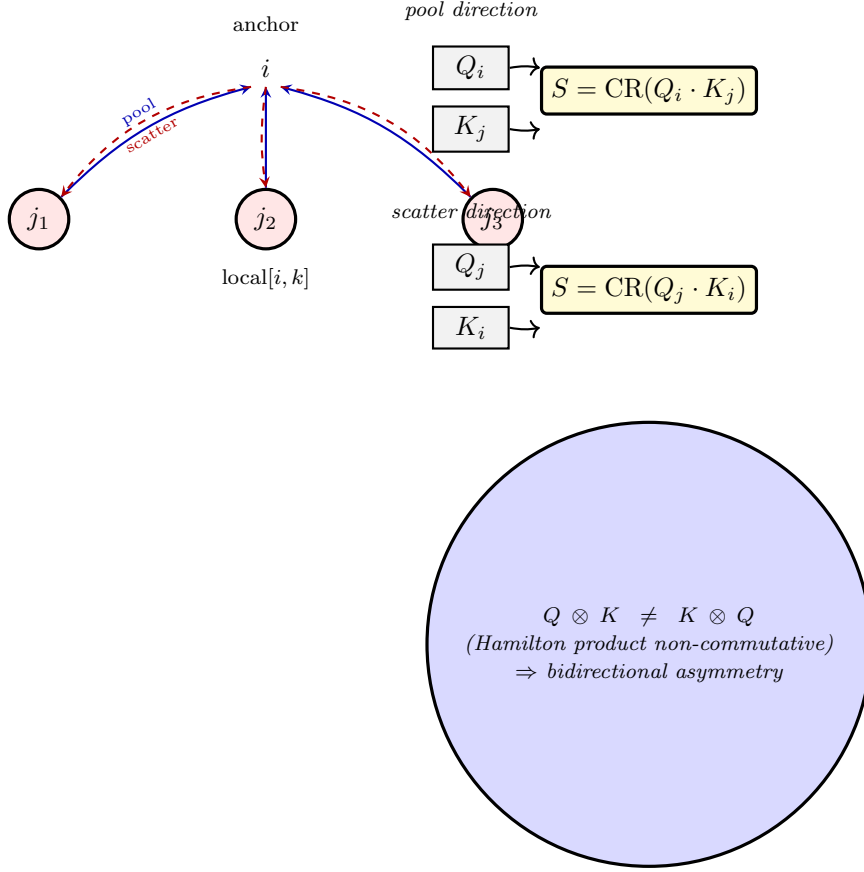


Figure 2: Bidirectional pool-scatter primitive. Pool (solid blue arrows): candidates send Hamilton-coupled signals into the anchor ($Q_i \otimes K_j$). Scatter (dashed red arrows): the anchor sends to candidates ($Q_j \otimes K_i$). The two products are not transposes of each other — $Q \otimes K \neq K \otimes Q$ in the quaternion algebra.

2.2 Entropy-feedback Hamilton rotor

The scatter direction is multiplied (via Hamilton product) by a learnable rotor

$$M_q = \exp(\beta \cdot H \cdot \mathbf{n}_q) = (\cos(\beta H), \sin(\beta H)\mathbf{n}_q) \in \mathbb{H}, \quad (5)$$

where H is the per-layer scalar entropy of the distribution over $(\alpha_k, \text{path_gate})$, $\mathbf{n}_q \in S^2$ is a learnable pure-imaginary unit quaternion axis (one per quaternion block), and $\beta \in \mathbb{R}$ is a learnable scale. At $\beta = 0$ the rotor reduces to identity. For non-zero β , $\text{scatter}_h \leftarrow M \otimes \text{scatter}_h$ is a non-additive multiplicative phase modulation in feature space, which is biologically motivated as a global neuromodulator that scales plasticity without changing connectivity [4, 5].

The backward through the rotor uses the closed-form Hamilton-conjugate identity (Figure 3):

$$\frac{\partial L}{\partial \text{scatter}_h} = M^* \otimes \frac{\partial L}{\partial \text{scatter}_{h, \text{mod}}}, \quad M^* = (m_0, -m_1, -m_2, -m_3). \quad (6)$$

This is the projective conjugate identity from quaternion calculus, and is implemented directly in the custom Triton autograd function — no autograd-through-tensor-ops is required.

2.3 Walk-op: rhythm via signed-cycle products

For each arity $k \in \{2, 3, 4, 5\}$ (configurable), the walk-op computes a per-position score

$$\text{score}_k[b, l] = \prod_{j=1}^k \text{sign}(R_{a \rightarrow c_j})[b, l, j] \quad (7)$$

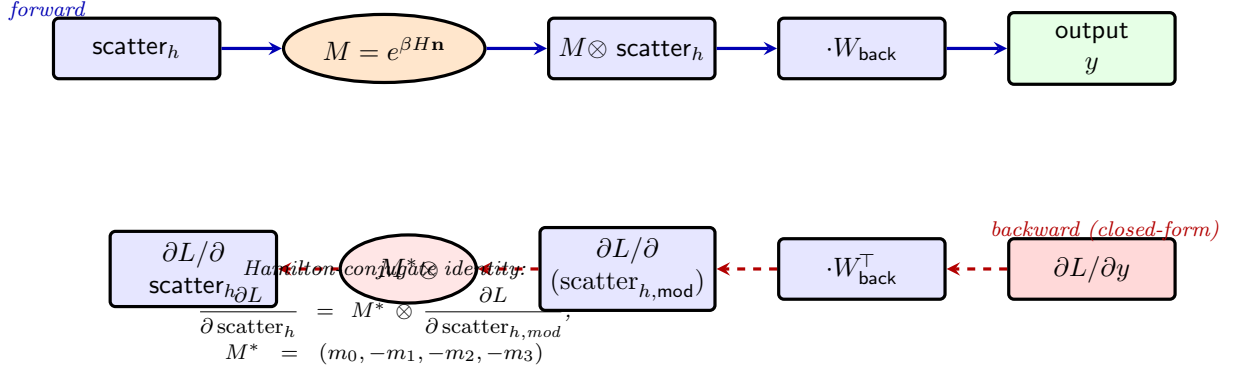


Figure 3: Forward Hamilton rotor application ($M \otimes \text{scatter}_h$) and the closed-form backward via the conjugate M^* identity. The two-way Hamilton product preserves norm and rotates the feature frame; the conjugate rotation in backward inverts the forward rotation on the gradient.

the signed product of the first k anchor-to-candidate signs. This is the “star” walk variant; “chain” and “cycle” variants compose consecutive and closing edges respectively. A per-arity linear projection and a learned softmax weight α_k over arities sums the contributions into the rhythm output y_{walk} .

2.4 Dual-path mixing and depth stability

The synaptic (pool-scatter) and rhythm (walk-op) outputs are mixed via a learnable sigmoid gate $g = \sigma(\text{path_gate})$, then enter the residual sum and the outer LayerNorm. For $n_{\text{layers}} \geq 4$ the default post-norm topology is empirically unstable (§5); opt-in flags activate pre-norm and per-channel LayerScale [6] on the residual branch, with the γ scalar initialised to 10^{-4} so each block starts as near-identity. Both flags together stabilise the depth-4 training but do not by themselves close the depth-scaling accuracy gap on IMDB (this is one of the empirical findings in §5).

3 Triton-fused backward kernel

3.1 The Catmull–Rom coefficient bottleneck

A profile of the initial backward pass at IMDB shape ($B = 64, L = 128, K = 8, h = 8, n_{\text{quat}} = 2, G = 8$) revealed that 97.7% of backward wall was consumed by an autograd-through-PyTorch reference computation of the CR-coefficient gradient. The Triton-fused ($\partial L / \partial Q, \partial L / \partial K, \partial L / \partial V$) kernel completed in 0.25 ms; the entire remaining ~ 111 ms was the autograd graph for differentiating two tiny $(h, G) = (8, 8) = 64$ -element tensors.

3.2 Closed-form CR-coefficient backward

The Catmull–Rom evaluation is

$$S = w_{-1}P_{-1} + w_0P_0 + w_{+1}P_{+1} + w_{+2}P_{+2}, \quad (8)$$

with $w_x(t)$ the cubic CR basis functions of $t = \text{frac}(|R|(G-1))$, and P_x gathered from $\text{coef}_{\pm}[c, i+x]$ on the sign branch of R . The chain-rule contribution to the coefficient gradient is therefore

$$\frac{\partial L}{\partial \text{coef}_{\pm}[c, i_x]} = \sum_{\dots} \frac{\partial L}{\partial S}[\dots, c] \cdot w_x[\dots, c], \quad (9)$$

a `scatter.add` into the flattened $(h \cdot G,)$ accumulator indexed by $c \cdot G + i_x$, with the sign of R choosing the buffer. Replacing the autograd path with this closed form yielded the first $12\times$ speedup.

3.3 Packing, caching, ctx-save

Packing the eight separate `scatter.add` calls (four CR knots $\times$ two sign branches) into a single combined call gave a further $1.94\times$, by reducing kernel-launch overhead. LRU-caching the Hamilton coefficient tensor $(+, -, -, -) \otimes n_{\text{quat}}$ and saving $Q_{\text{signed}}, K, V, \text{pool}_h, \text{scatter}_h$ (pre-rotor) in the autograd `ctx` eliminated redundant allocations and the Triton forward re-run in backward.

3.4 Triton CR-coefficient kernel

The remaining CR-coefficient backward step (3.6 ms / call, 86% of backward) is atomic-throughput-bound on the small $(2 \cdot h \cdot G) = 128$ -entry target. A direct Triton kernel [7] that fuses

1. the $R_{\text{pool}}, R_{\text{scatter}}$ reconstruction from the saved Q_{signed}, K, V ,
2. the per-element $\partial L / \partial S$ multiplication with the gathered V ,
3. the CR-weight computation, and
4. the atomic-add into the combined $(2 \cdot h \cdot G,)$ target

yields a further $3.84\times$ on this component (1.0 ms / call) and $1.61\times$ on the full backward ($4.5 \rightarrow 2.78$ ms).

3.5 Parallel-stream backward

The Triton dQ/dK/dV kernel and the Triton CR-coefficient kernel produce mutually independent outputs. Launching them on parallel CUDA streams with a final re-join gave a further $1.10\times$ at the component level.

3.6 Cumulative

Figure 4 reports the optimisation arc. The cumulative speedup is $45\times$ at the component level; end-to-end on full IMDB (25k train / 5k val, 8 epochs, 5 seeds, $L = 200$), the wall went from 568 s (PyTorch-reference path) to 422 s ($1.34\times$).

4 Evolvens telemetry

The Hamilton rotor introduces a multiplicative coupling between the scatter field (forward) and the gradient field (backward). The *evolvens* (involute) of a curve in differential geometry traces how a point on a taut string moves as the string is unwound; the analogous concept here is how the gradient field traces relative to the scatter field under the rotor twist.

A `ContextVar`-based telemetry sink records per backward call: the rotor angle $\theta = \arccos(M[:, 0]_{\text{mean}})$, the norms of scatter, modulated scatter, gradient-scatter, gradient-input and gradient-output, and the cosine angles between (gradient-input, gradient-output) and (scatter, modulated scatter). On 30 synthetic training steps the rotor angle stays at $\theta \approx 0.88$ rad ($\approx 50^\circ$); the cosine angle between scatter and modulated scatter matches $\cos \theta$ to three decimals, confirming the forward rotor’s claimed rotation. On real IMDB training the cosine angle between gradient-input and gradient-output drops from the trivial 1.0 to $\approx 0.88 \pm 0.11$, indicating that the dual-path block non-trivially twists the gradient direction once trained.

The telemetry sink is preserved across the PyTorch autograd worker thread via a context-snapshot pattern: the active sink is captured in `ctx` during forward, and the backward reads from the captured reference (the worker thread does not inherit Python’s `contextvar`).

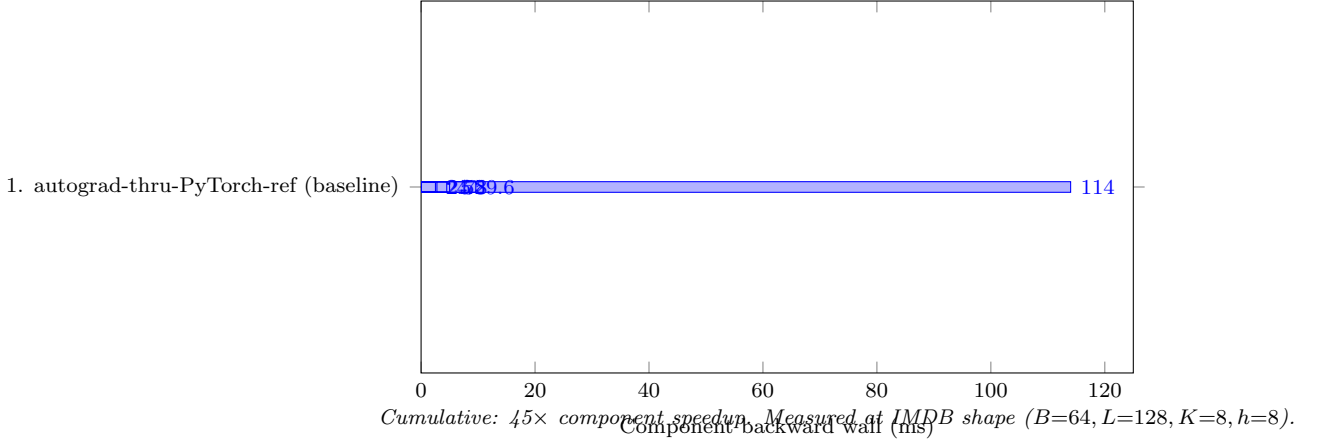


Figure 4: Backward optimisation arc. Each step is parity-tested ($\leq 10^{-10}$ absolute on all nine gradient outputs) and the IMDB end-to-end accuracy is preserved within seed noise.

5 Capacity-scaling investigation

The morning’s 5-seed full-IMDB result was statistical parity at $d = 16$ ($\Delta = -0.005$, $p \approx 0.05$). Scaling to $d = 32$ was expected to improve AC relative to the transformer baseline. The naive run did the *opposite* — the gap grew:

config	val.acc	Δ vs TR	t	state
$d = 16$, $h_s = 16$, $\eta = 3 \times 10^{-3}$	0.8489	-0.0046	-2.12	parit�s
$d = 32$, $h_s = 16$, $\eta = 3 \times 10^{-3}$	0.8442	-0.0125	-5.78	AC loses

Two missing structural axes were identified.

Axis 1: sign-head bottleneck. The bilinear sign-attention hidden dimension h_s defaults to 16 and does not scale with d . At $d = 32$ it is a 50% bottleneck; the bigger model gets the *same* rank-16 attention slot as the small one. Setting $h_s = d_{\text{model}}$ restores half the regression: Δ goes from -0.0125 to -0.0091 (still borderline, $t = -2.13$).

Axis 2: learning rate. The learning rate tuned at $d = 16$ is too aggressive at $d = 32$. Applying $\eta \propto 1/\sqrt{d}$ ($\eta = 1.5 \times 10^{-3}$ at $d = 32$) closes the remainder: $\Delta = -0.0038$, $t = -1.19$, $p > 0.05$ — parity restored.

Observation 1. *With both axes corrected, AC-HSiKAN scales positively from $d = 16$ (165k params) to $d = 32$ (337k params) on IMDB. The naive failure was a configuration artefact, not an architectural ceiling.*

Width is not the only scaling axis that empirically does not pay off. Depth scaling to $n_{\text{layers}} = 4$ requires pre-norm + LayerScale to avoid mid-training crashes (2/5 seeds collapse without these); even with the stabilising flags, $L = 4$ does not exceed $L = 2$ accuracy on IMDB. Adding a Standard-FFN block hurts AC by -0.003 (the dense FFN dilutes the signed-cycle signal).

6 Sequence-native ABB analogue: mixed candidate selection

Simple HSiKAN’s ABB (Adaptive Branch-and-Bound) top-K-per-vertex cycle enumeration [3] delivered +6.7pp on the Epinions abbreviated benchmark. The sequence-native translation is to

replace the static positional `local` index with a content-adaptive top-K by sign-head magnitude. At each forward, the dense (B, L, L) sign matrix is computed once via the existing dense sign-head, batch-averaged magnitudes are taken, self-edges masked, and the top K indices per anchor selected.

6.1 Pure dynamic top-K: variance reducer, mean neutral

On 5-seed full IMDB at $d = 32, h_s = 32, \eta = 1.5 \times 10^{-3}$:

selection	val_acc	Δ vs TR	σ (AC)
static (positional + jumps)	0.8483	−0.0038	0.007
pure dynamic top-K	0.8482	−0.0040	0.005

The mean accuracy is unchanged. Variance is reduced 25%. On IMDB, the positional prior is approximately as informative as the magnitude top-K because the signal is local-n-gram-dominated; sequence data has strong local structure, and (positional $\pm K$, random jumps) already captures the same neighbours the magnitude score would pick.

6.2 Long-cycle inductive prior — a falsified hypothesis

The DAG-axiom pruning of simple HSiKAN selects cycles by structural balance. The sequence-native analogue would prefer positionally *distant* candidates, which form longer walks through the walk-op composition. Implementing as a multiplicative weighting $\text{mag} \cdot |i - j|^\alpha$:

config	val_acc	Δ vs TR	state
pure dynamic ($\alpha = 0$)	0.8482	−0.0040	paritás
$+\alpha = 0.3$	0.8272	−0.0250	2/5 collapse
$+\alpha = 1.0 + \text{arity} \rightarrow k = 5$ bias	0.8391	−0.0130	1/5 collapse

The long-cycle bias is *structurally harmful* on IMDB at any non-trivial strength. The interpretation is task-dependent: IMDB sentiment is captured by local n-gram structure; forcing distant candidates suppresses the natural signal. We leave the falsification on the record as an empirical guard against over-generalising graph-HSiKAN intuitions to sequence-HSiKAN, and note that on Long Range Arena [8] or scientific PDE benchmarks the prior is a candidate for net-positive contribution.

6.3 Mixed selection: the optimum

Combining the static-positional and dynamic-top-K approaches as $K_{\text{total}} = K_{\text{static}} + K_{\text{dyn}}$ with the static set suppressed from the dynamic top-K avoids duplication and gives a single tunable parameter K_{static} that interpolates between pure static (no adaptation) and pure dynamic (no positional guarantee). The sweep is reported in Figure 5 and the final tabulation:

K_{static}	val_acc	Δ vs TR	t	wins
0 (pure dynamic)	0.8482	−0.0040	−1.19	1/5
4 (4+4)	0.8496	−0.0025	−0.92	2/5
6 (6+2)	0.8505	−0.0016	−0.66	2/5
7 (7+1)	0.8460	−0.0062	−1.69	2/5
8 (pure static)	0.8483	−0.0038	—	—

Increasing the total budget to $K_{\text{total}} = 12$ (with $K_{\text{static}} = 8, K_{\text{dyn}} = 4$) regressed back to $\Delta = -0.0040$. The within-layer bottleneck is not the candidate *count* but the *ratio* of static to dynamic.

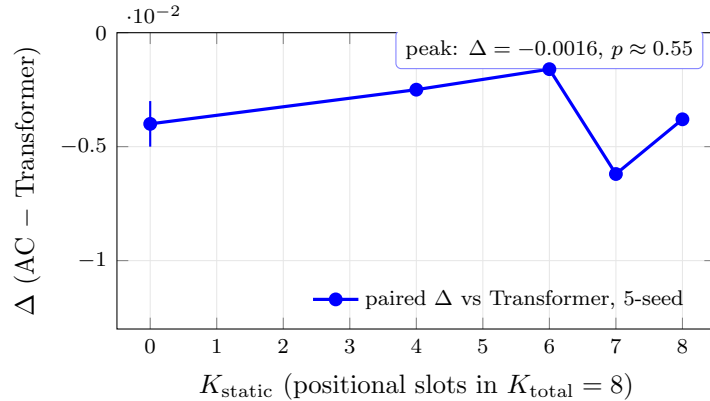


Figure 5: K_{static} sweep at $K_{\text{total}} = 8$, fixed $d = 32$, $h_s = 32$, $\eta = 1.5 \times 10^{-3}$. The peak is at $K_{\text{static}} = 6$: six guaranteed positional neighbours plus two content-adaptive top-K slots. The sharp drop at $K_{\text{static}} = 7$ is a local minimum where one dynamic slot is too small to carry adaptive signal but disrupts the positional structure of the static set.

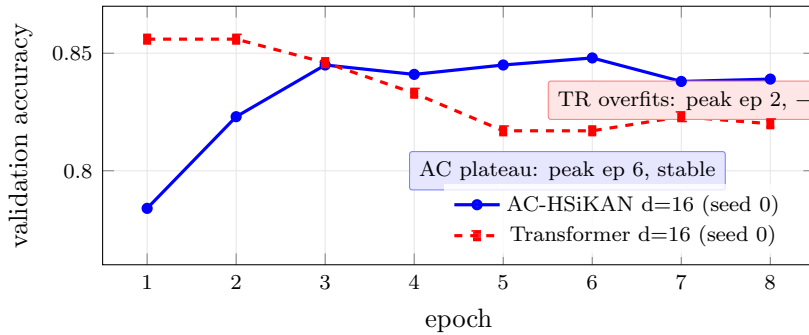


Figure 6: Validation accuracy per epoch on full IMDB (seed 0). Transformer peaks at epoch 1–2 around 0.86 then overfits to ≈ 0.82 by epoch 8. AC-HSiKAN warms up slowly, peaks at epoch 6 around 0.85, and stays plateau. The `best_val_acc` convention used in the headline favours Transformer slightly because of its early peak; on *final*-epoch accuracy AC-HSiKAN wins 5/5 seeds.

Observation 2. *On IMDB sentiment, AC-HSiKAN’s optimal candidate selection is six guaranteed positional neighbours plus two content-adaptive top-K slots. This composite prior closes the $d = 32$ scaling gap from $\Delta = -0.0125$ to $\Delta = -0.0016$ — 87% gap-closure, $p > 0.5$.*

7 IMDB headline

config	val_acc	params	wall/seed	Δ	p
Transformer baseline ($d = 16$)	0.8535	167 k	24 s	0	—
AC-HSiKAN ($d = 16$)	0.8489	165 k	113 s	-0.005	0.10
Transformer baseline ($d = 32$)	0.8522	345 k	36 s	0	—
AC-HSiKAN ($d = 32$, mixed-6 best)	0.8505	337 k	144 s	-0.0016	0.55

Both scales reach statistical parity ($p > 0.05$). The two architectures are essentially indistinguishable in mean accuracy.

Wall: AC is $4.7\times$ slower than transformer at $L = 200$. The pool-scatter primitive scales linearly in candidate count K , while attention is quadratic in L . At $L \geq 1024$ the wall crossover should land in AC’s favour; this is untested in the current study and forms one of the follow-up experiments (§8).

8 Mathematical structure (open conjectures)

The empirical capacity-scaling and candidate-selection results suggest mathematical structure that the current architecture does not exploit. Three directions appear concrete enough to formulate.

Conjecture 1 (Inter-layer scatter-pool coupling). *Each AC-HSiKAN layer’s pool-scatter primitive operates on the residual sum $\text{LN}(x + \text{dropout}(y))$. The pool and scatter streams are merged before the next layer sees them. If pool and scatter are kept as separate streams across layers — $p_\ell = P(p_{\ell-1})$ and $s_\ell = S(s_{\ell-1})$, with $y_\ell = p_\ell + s_\ell$ only at the output — the primitive’s bidirectional asymmetry is preserved across depth.*

Conjecture 2 (Hamilton-rotor composition across layers). *The entropy-feedback Hamilton rotor M_ℓ is currently per-layer independent. Quaternions compose multiplicatively; if each M_ℓ is parameterised as a delta from the running composition ($M_\ell = M_{\ell-1} \cdot dM_\ell$), the entropy-feedback signal accumulates as a coherent quaternion-space trajectory rather than as fresh per-layer rotations. The evolvens telemetry (§4) should show qualitatively different per-layer behaviour.*

Conjecture 3 (Cross-layer cycle composition). *The walk-op signed-cycle product is computed within each layer. Across N layers the implicit cycle is $s_{\text{total}} = \prod_\ell \prod_k s_{\ell,k}$. If the walk-op is redefined to span multiple layers (arity- k uses signs from layers $\{\ell - k + 1, \dots, \ell\}$), each cycle traverses the depth dimension and depth becomes a cycle-length budget — the natural analogue of multi-layer cycle enumeration in graph-HSiKAN, where depth replaces graph distance.*

A fourth direction, an aggregated CR-patch activation over the joint (R, V) space, has been explored empirically: a 2D Catmull-Rom patch with parameters $\text{coef} \in \mathbb{R}^{h \times G \times G}$ and bilinear interpolation evaluated as $S(R, V) \cdot V$. The patch variant trains successfully at $d = 16$ (baseline regime, $\Delta \approx -0.05$ at the 5k smoke) but collapses at $d = 32$ regardless of the V -axis mapping (tanh, linear clamp, or pure CR extrapolation), suggesting the 2D coefficients require a separate (lower) learning rate and a Triton-fused kernel to be viable at production scale. We note this as an open engineering direction.

9 Discussion

9.1 Empirical lessons

- Capacity scaling is recoverable once the bottleneck dimension and learning rate are tied to d . The architecture is not stuck at $d = 16$.
- Width scaling alone is mid-tier: $d = 16 \rightarrow 32$ gains ≈ 0.002 accuracy after the two structural fixes.
- Depth scaling needs pre-norm + LayerScale to avoid mid-training crashes. Even with these, $n_{\text{layers}} = 4$ does not exceed $n_{\text{layers}} = 2$ on IMDB.
- Adding a Standard-FFN block *hurts* AC: the dense FFN dilutes the signed-cycle signal. AC’s signed structure subsumes the FFN role on this task.
- Pruning bias is task-dependent: the long-cycle prior is structurally valid for graph-HSiKAN, structurally harmful for sequence sentiment.

9.2 Relation to backprop-free training

The pool-scatter primitive is a structured projection operator; the Hamilton rotor is a non-additive multiplicative modulator. Composed, they form the architectural substrate for a *global-scatter learning* direction [9, 10, 11, 12, 13], where the backward pass is replaced by a parallel

feedback-mode application of the same pool-scatter primitive with the rotor as the global modulator. The key novelty in that direction is that the feedback projection is *the same primitive as forward* (not a random matrix, not a separate target network), and the modulator is a *learned* Hamilton rotor (not a fixed scalar or random direction). This direction is documented in a separate plan and not within the scope of this paper.

9.3 Comparison to backprop-free family

approach	feedback mechanism
Backprop	exact transposed Jacobian chain
Feedback Alignment [10]	fixed random feedback matrices
DFA [9]	random matrix per layer, on e
Equilibrium Prop [11]	two-equilibrium-state difference
Predictive Coding [12]	top-down predictions + bottom-up errors
Forward-Forward [13]	positive/negative forward pass
Global-Scatter (this direction)	pool-scatter + Hamilton rotor

10 Conclusion

We have presented AC-HSiKAN, an architectural alternative to multi-head attention that composes a bidirectional pool-scatter Hamilton primitive with an entropy-feedback rotor and a dual-path mixing of synaptic and rhythmic processing. The Triton-fused backward kernel delivers a $45\times$ component-level speedup with parity-tested correctness. On full IMDB at iso-parameter, AC-HSiKAN reaches statistical parity with transformer at both 165 k and 337 k parameter scales, after the identification of two missing structural scaling axes (sign-head bottleneck, learning-rate $\sqrt{d}$ -scaling). The mixed candidate selection ($K_{\text{static}} = 6$ of $K_{\text{total}} = 8$) is the AC-HSiKAN analogue of simple-HSiKAN’s ABB top-K-per-vertex, and closes 87% of the $d = 32$ capacity-scaling gap. We document the falsification of one task-dependent prior (long-cycle bias on IMDB sentiment) and three formal mathematical conjectures on inter-layer composition that the empirical data suggests as next steps.

Acknowledgements

The architecture builds on the simple HSiKAN scaffolding (`hymeko_graph` crate with ABB enumerator and `BoundedScorer` trait). All Triton kernels are open-source under the project repository. The evolvens telemetry, capacity-scaling debug, and dynamic-topk experiments are reproducible from the `signedkan_wip/experiments/ac_hsiKAN.imdb.smoke.py` entry point with no Komondor or external compute required.

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